8. Insertion Sort with Duplicate Elements

Aim

To write a C program to implement **Insertion Sort** on an array containing duplicate elements and display the sorted array.

Procedure

1. Create an array containing duplicate elements.
2. Consider the first element as the sorted portion.
3. Select the next element as the `key`.
4. Compare the key with the elements before it.
5. Shift elements greater than the key one position to the right.
6. Insert the key into its correct position.
7. Repeat the process until all elements are sorted.
8. Display the sorted array.

main.c

Run ▶

Output

```c
#include <stdio.h>
void insertionSort(int arr[], int n)
{
    int i, j, key;
    for (i = 1; i < n; i++)
    {
        key = arr[i];
        j = i - 1;
        while (j >= 0 && arr[j] > key)
        {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = key;
    }
}
void display(int arr[], int n)
{
    int i;
    for (i = 0; i < n; i++)
    {
        printf("%d ", arr[i]);
```

```
Array Before Sorting: 5 2 8 2 9 5 1 2
Array After Sorting:  1 2 2 2 5 5 8 9
```

main.c

Run ▶

Output

```c
    }
}
void display(int arr[], int n)
{
    int i;
    for (i = 0; i < n; i++)
    {
        printf("%d ", arr[i]);
    }
    printf("\n");
}
int main()
{
    int arr[] = {5, 2, 8, 2, 9, 5, 1, 2};
    int n = 8;
    printf("Array Before Sorting: ");
    display(arr, n);
    insertionSort(arr, n);
    printf("Array After Sorting:  ");
    display(arr, n);
    return 0;
}
```

```
Array Before Sorting: 5 2 8 2 9 5 1 2
Array After Sorting:  1 2 2 2 5 5 8 9
```

Result

Thus, the C program successfully sorts an array containing **duplicate elements** using the Insertion Sort algorithm.